Information Systems Applications

Chapter 1: Introduction to Information Systems

By the end of this lecture, you should be able to:

- Define Information Systems (IS).
- Explain the difference between Data and Information.
- Identify the main components of an Information System.
- Explain how an Information System works.
- Distinguish between IS, IT, and Computer Science.
- Identify major types of Information Systems.
- Identify common applications of Information Systems.

What is an Information System?

An **Information System (IS)** is a set of interconnected components that:

- Collect data
- Process data
- Store data
- Distribute information

Information Systems support:

- Decision-making
- Coordination
- Control
- Analysis
- Organizational activities

Why Do We Need Information Systems?

Organizations use Information Systems to:

- Collect and organize data.
- Transform data into useful information.
- Support decision-making.
- Improve efficiency and productivity.
- Support communication and coordination.
- Improve services and organizational performance.

Data vs. Information

Data

Raw facts and figures that have not yet been processed.

Information

Processed data that has meaning and context.

Data → Processing → Information

Example:

80, 75, 90, 65 → **Data**

Average = 77.5% → **Information**

Components of an Information System

An Information System consists of:

1. **Hardware**
2. **Software**
3. **Data**
4. **People**
5. **Procedures / Processes**

An Information System is not just computers and software.

Hardware

Hardware consists of the physical devices used by an Information System.

Examples:

- Computers
- Servers
- Storage devices
- Network devices
- Sensors
- Mobile devices

Software

Software provides the instructions and applications used to process data.

Examples:

- Operating Systems
- Database Management Systems (DBMS)
- Business Applications
- Web Applications
- Mobile Applications

Data

Data is the raw material processed by an Information System.

Examples:

- Student ID
- Student Name
- Grades
- Course Code
- Customer Information
- Sales Transactions

People

People are an essential component of an Information System.

Examples:

- End Users
- Managers
- System Analysts
- Database Administrators
- IT / IS Professionals

Procedures and Processes

Procedures define how activities are performed within the system.

Example: University Course Registration

Login → Select Courses → Validate → Confirm Registration

Processes connect **people and technology** to achieve a specific result.

How Does an Information System Work?

Input → Processing → Output

Input
Data entered or captured by the system.

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Processing
Data is transformed, calculated, classified, or analyzed.

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Output
Useful information is produced.

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Feedback
Results can be reviewed and used to improve the process.

IS Operations

Information Systems perform several important operations:

- **Data Capture** – Collect data.
- **Data Processing** – Transform data.
- **Data Storage** – Save data.
- **Data Retrieval** – Access stored data.
- **Data Dissemination** – Deliver information to users.

IS, IT, and Computer Science

Computer Science

Focuses on computing concepts, algorithms, software, and computational systems.

Information Technology (IT)

Focuses on implementing, maintaining, and supporting technology.

Information Systems (IS)

Focuses on using **technology, people, data, and processes** to support information and organizational needs.

Major Types of Information Systems

At a high level, organizations use different types of Information Systems:

- Transaction Processing Systems (TPS)
- Management Information Systems (MIS)
- Decision Support Systems (DSS)
- Executive Information / Support Systems (EIS / ESS)
- Enterprise Information Systems

Applications of Information Systems

Information Systems are used in many industries:

- 🏢 Banking and Financial Services
- 🏥 Healthcare
- 🎓 Education
- 🛒 E-Commerce
- 🏛️ Government
- 🚚 Transportation and Logistics

- 🏭 Manufacturing
- 💼 Business

Benefits of Information Systems

Information Systems can provide:

- Improved efficiency
- Better decision-making
- Faster access to information
- Improved communication
- Better coordination
- Better customer service
- Reduced time and operational costs

IS and Emerging Technologies

Modern Information Systems increasingly integrate:

- **Artificial Intelligence (AI)**
- **Machine Learning (ML)**
- **Cloud Computing**
- **Big Data Analytics**
- **Internet of Things (IoT)**
- **Blockchain**

These technologies can make Information Systems more **intelligent, connected, and adaptive**.

Real-Life Example

University Information System

Input

- Student information
- Courses
- Grades

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Processing

- Course registration
- GPA calculation
- Academic evaluation

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Output

- Transcript
- GPA
- Registration status

Users

- Students
- Instructors
- Administrators